

SAI ROHAN

Sr.Data Engineer

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Professional Summary:

- Around 8+ years of IT experience with Data Engineer/Analyst and coding with analytical programming using SQL, Python, Snowflake and AWS.
- Good working knowledge in multi-tiered distributed environment, good understanding of Software Development Lifecycle (SDLC)-Agile and Waterfall Methodologies
- The ability to develop reliable, maintainable, efficient code in most of SQL, Hive and Python.
- Experience in importing and exporting data using Sqoop from HDFS to Relational Database Management Systems (RDBMS) and from RDBMS to HDFS.
- Strong knowledge of various data warehousing methodologies and data modeling concepts.
- Experience in the development of ETL processes and frameworks for large-scale, complex datasets
- Experience with application development on Linux, python, RDBMS, NoSQL and ETL solutions
- Experience designing and operating very large Data Warehouses.
- Experience in the development of ETL processes and frameworks like matillion for large-scale, complex datasets
- Good knowledge in BI and data visualization tools such as Tableau
- Expertise in writing Hadoop Jobs to analyze data using MapReduce, Hive, Pig.
- Gained Knowledge to write cloud formation templates and deployed AWS resourcing.
- Design and build data processing pipelines using tools and frameworks.
- Understanding data transformation and translation requirements and which tools to leverage to get the job done.
- Understanding data pipelines and modern ways of automating data pipelines using cloud-based techniques.
- Good understanding and hands on experience in setting up and maintaining NoSQL Databases like MongoDB, and HBase
- Familiarity with new advances in the data engineering space such as EMR and NoSQL technologies like DynamoDB.
- Good knowledge in data visualization tools such as Tableau
- Ability to work on several python packages like Pandas, NumPy, Matplotlib, Beautiful Soup, Pyspark
- Hands-on experience in working with Continuous Integration and Deployment (CI/CD).
- Defining user stories and driving the agile board in JIRA during project execution, participate in sprint demo and retrospective.
- Strong ability to compile Java programming including OOPS concepts, Class, Method, inheritance, Encapsulation, Loop, Exception handling, etc.
- Hands-on experience in application development using Java, RDBMS (SQL), and Linux/Unix shell scripting.
- Experience in using various version control systems like Git.
- Strong analytical skills with the ability to collect, organize and analyze large amounts of information with attention to detail and accuracy.
- Possess good interpersonal, analytical presentation Skills, ability to work in Self-managed and Team environments.

Technical Skills:

Programming Languages	Python,Java, SQL, PHP, C++, Shell Scripting
SDLC Methodologies	Agile/SCRUM, Waterfall
Operating Systems	Windows, Linux, Unix
Python Libraries	Pyspark, Pandas, Beautiful Soup, Jinja2, NumPy, SciPy, Matplotlib, Unit test
Big Data Tools	Hadoop3.3, Hive3.2.1, Kafka2.8, Scala, MapReduce, Sqoop, Apache flink
Cloud Tools	Azure and AWS (S3, RDS, Dynamo, EMR, Redshift, Glue),
Databases	MS SQL Server, MySQL, PostgreSQL, Oracle, MongoDB, SQLite, Dynamo DB
Version Controls	Git, GitHub, Bitbucket

Other Tools	Snowflake, Databricks, Hive Spark, Matillion, ETL, JIRA, Docker, MS Excel, Data Pipeline, Data Modeling
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Professional Experience:

Client: Vanguard

May 2023 to Present

Role: Sr. Data Engineer

Responsibilities:

- Experienced in development using Cloudera Distribution System.
- Design and develop ETL integration patterns using Python on Spark.
- Optimize the PySpark jobs to run on Secured Clusters for faster data processing.
- Used Python for SQL/CRUD operations in DB, file extraction/transformation/generation.
- Developed spark applications in Python (PySpark) on a distributed environment to load huge numbers of CSV files with different schema into Hive ORC tables.
- Designing and Developing Apache NiFi jobs to get the files from transaction systems into data lake raw zone.
- Analyzed the user requirements and implemented the use cases using Apache NiFi.
- Migrating an entire oracle database to BigQuery and using DAG and composers.
- Build data pipelines in airflow in GCP for ETL related jobs using different airflow operators.
- Experience in GCP Dataproc, GCS, Cloud functions, BigQuery.
- Working on Google cloud storage buckets to store the project specific files and implement them together.
- Coordinated with team and Developed framework to generate Daily adhoc reports and Extracts from enterprise data from BigQuery.
- Used cloud shell SDK in GCP to configure the services Data Proc, Storage, BigQuery and DQF
- Performed Data Quality checks and maintained the file data throughout the process from preprocessing to big Query loading.
- Resolved several issues when trying to load the data into the Google cloud.
- Worked on Jenkins for creating the tables based on build parameters and deploying tables from dev to QA and Prod
- Worked on reading and writing multiple data formats like JSON, ORC, Parquet on HDFS using PySpark.
- Experienced with Azure Microservices, Azure Functions and Azure Solutions.
- Hands on experience on Azure VPN-Point to Site, Virtual networks, Azure Custom security, endpoint security and firewall.
- Caching, SQL Azure, NoSQL, Storage, Network services, Azure Active Directory, API Management, Scheduling, Auto Scaling, and PowerShell Automation.
- Involved in developing the Azure Solution and Services like IaaS and PaaS.
- As a Hadoop Developer, my role is to manage the Data Pipelines and Data Lake.
- Have experience of working on Snow -flake data warehouse and have hands-on experience working with SnowSQL.
- Developed Spark code using Scala and Spark-SQL/Streaming for faster processing of data.
- Designed a custom Spark REPL application to handle similar datasets.
- Used Hadoop scripts for HDFS (Hadoop File System) data loading and manipulation.
- Performed Hive test queries on local sample files and HDFS files.
- Worked on creating Azure Data Factory and managing policies for Data Factory and Utilized Blob storage for storage and backup on Azure. Developed multiple Map Reduce jobs in Java for data cleaning and preprocessing.
- Developed Spark Applications by using Scala, and Java and Implemented Apache Spark data processing project to handle data from various RDBMS and Streaming sources.
- Developed the application on Eclipse IDE.
- Worked on Configuration of Internal load balancer, load balanced sets and Azure Traffic manager.
- Developed Hive queries to analyze data and generate results.
- Used Spark Streaming to divide streaming data into batches as an input to Spark Engine for batch processing.
- Worked on analyzing Hadoop clusters and different Big Data analytic tools including Pig, hive, HBase, Spark and Sqoop.

- Exported data from HDFS to RDBMS via Sqoop for Business Intelligence, visualization, and user report generation.
- Used Scala to write code for all Spark use cases.
- Analyzed user request patterns and implemented various performance optimization measures including implementing partitions and buckets in HiveQL.
- Caching, SQL Azure, NoSQL, Storage, Network services, Azure Active Directory, API Management, Scheduling, Auto Scaling, and PowerShell Automation.
- Assigned name to each of the columns using the case class option in Scala.
- Worked on migrating Map Reduce programs into Spark transformations using Spark and Scala, initially done using Python (PySpark).
- Involved in converting the HQL's into spark transformations using spark RDD with support of Python and Scala.
- Developed multiple Spark SQL jobs for data cleaning.
- Created Hive tables and worked on them using Hive QL.
- Experienced on Big data with Azure. Connecting HDInsight to Azure and working on Big Data technology.
- Assisted in loading large sets of data (Structure, Semi Structured, and Unstructured) to HDFS.
- Developed Spark SQL to load tables into HDFS to run select queries on top.
- Developed analytical components using Scala, Spark, and Spark Stream.
- Used Visualization tools such as Power view for excel, Tableau for visualizing and generating reports.
- Worked on the NoSQL databases HBase and mongo DB.
- Worked on Configuration of Internal load balancer, load balanced sets and Azure Traffic manager.

Client: Deutsche bank, Jacksonville, FL

Mar 2022–Apr 2023

Role: Sr. Data Engineer

Responsibilities:

- Tuned existing Pyspark applications to load billion records everyday into Hadoop from RDBMS.
- Apply Hadoop technologies for responsive front-end experience by creating HBase/Phoenix tables over HIVE tables based on the business requirement.
- Developed data pipelines to ingest, process, and analyze data from various sources including relational databases and web services.
- Designed and deployed AWS EMR clusters for processing large datasets, improving data processing speed.
- Designed and implemented scalable data warehousing solutions in Snowflake, enabling efficient storage and querying of large datasets across multiple business units.
- Developed and optimized SQL queries in Snowflake to enhance data retrieval performance, reducing query execution times by 50%.
- Integrated Snowflake with ETL tools like Apache Airflow and AWS Glue, automating data pipelines and improving data flow between cloud environments.
- Created and maintained data pipelines using AWS Glue, automating data integration tasks.
- Orchestrated data workflows with AWS Airflow, improving job execution and monitoring.
- Developed Spark Programs using Scala and Java APIs and performed transformations and actions on RDDs.
- Developed UDFs in Java for hive and pig and worked on reading multiple data formats on HDFS using Scala.
- Developed code to obtain data from the System of Record and establish data feed for the data analysis automated.
- Developed Data Mapping, Data Governance, Transformation and Cleansing rules for the Master Data Management Architecture involving OLTP, ODS and OLAP
- Engineered data pipelines using python spark, handling large-scale data processing tasks and ensuring data accuracy.
- Worked extensively with AWS Glue ETL and Lambda functions for real-time data processing and event-driven architectures.
- Utilized AWS Lake Formation and S3 to create secure, scalable data storage solutions, enabling efficient data lake management.
- Optimized data serialization and deserialization processes with formats such as Iceberg and JSON-LD, improving data handling capabilities.

- Contributed to DevOps practices, including CI/CD automation and TDD, to maintain high-quality code and efficient deployment cycles.
- Leveraged in-depth understanding of end-to-end customer experience to identify pain points and latent customer needs.
- Wrote entities in Scala and Java along with named queries to interact with the database.
- Developed prototype Spark applications using Spark-Core, Spark SQL, and Data Frame API and developed several custom User-defined functions in Hive & Pig using Java & python
- Mentored teams, leading them to be self-directed with strong accountability, decision making conflict resolution, and transparency.
- Supported the development of cloud-based analytics platforms, enabling real-time data insights.
- Designed and implemented data integration solutions using Informatica, Scala spark, and AWS services.
- Managed and optimized relational and NoSQL databases, including Oracle and DynamoDB.
- Implemented RESTful APIs and server-side logic using Node.js and Express.js, facilitating seamless communication between frontend Angular applications and backend data processing systems, ensuring data integrity and security.
- Build complex SQL queries to get the circuit information from the OpenWay and different databases.
- Developed Scala code to handle the meters, circuit information for the business use case.
- Developed and maintained ETL processes with AWS Glue, ensuring data consistency and reliability.
- Managed workflow automation using AWS Airflow, significantly improving task automation and monitoring.
- Provisioned and managed infrastructure with Terraform, enhancing deployment efficiency and consistency.
- Integrated Cloudera Hadoop clusters for enterprise data warehousing, supporting data-driven decision-making.
- Worked closely with data scientists and analysts to deliver cloud-based analytics solutions, enhancing business reporting capabilities.
- Collaborate with application developers, database architects, data analysts and data scientists to ensure optimal data delivery architecture throughout ongoing projects/operations.
- Create tables in HIVE similar to DB tables using star schema, as per the business request.
- Stayed abreast of latest trends and advancements in web development technologies, frameworks, and tools, continuously learning and applying new techniques to enhance productivity, performance, and innovation in data-driven web applications.

Client: Xcel Energy, Minneapolis, MN

Oct 2020 – Mar 2022

Role: Sr. Data Engineer

Responsibilities:

- Working as a **Data Engineer**, Assisted in leading the plan, building, and running states within the Enterprise Analytics Team.
- Worked in an Agile **development** environment in sprint cycles of two weeks by dividing and organizing tasks.
- Highly Involved into **Data Architecture** and Application Design using **Cloud and Big Data solutions** on AWS.
- Work independently to develop and test **Hive** based blocks to load and register data into **Data Lake** and **Data Warehouse**.
- Migrated application from internal data center to **AWS cloud platform**.
- Used and developed python scripts to migrate data to AWS.
- Involved in **Dataset Registration** in Exchange for **S3, One Lake, and Snowflake**.
- Created ETL jobs using **matillion** to load server data into the **Snowflake Data Warehouse**.
- Worked on **Data management** and **Data Integration**.
- Implemented solutions for ingesting data from various sources and processing the Data-at-Rest utilizing **Big Data technologies** such as **Hadoop, Map Reduce** and **Hive**.
- Provided guidance to the development team working on **PySpark** as **ETL platform**.
- Developed **data pipeline** using **Sqoop** to ingest cargo data and customer histories into **HDFS** for analysis.
- Maintain existing **ETL workflows**, data management and data query components.
- Extensively worked using **AWS services** along with wide and in depth understanding of each one of them
- Involved in creating **Hive tables** and loading and analyzing data using **hive queries**.
- Built **REST APIs** to easily add new analytics or issuers into the model.

- Developed **Map Reduce** program for parsing and loading into **HDFS**.
- Used **Hive** to do transformations, event joins and some pre-aggregations before storing and registering data into **snowflake**.
- Developed **UDFs** in **Java** as and when necessary to use in **Pig** and **HIVE** queries.
- Used **Hive SQL** to do analysis on the data and identify different correlations.
- Used **Hive** script and **Amazon Elastic MapReduce** for **unit testing**.
- Imported data from **DynamoDB** to **Redshift** in Batches using **Amazon Batch** using TWS scheduler.
- Performed **Data analysis** and **Data validation** on **Snowflake** and **Hue Editor**.
- Capacity planned for cloud infrastructure with **AWS console**, **EC2 Instances**, **AWS EMR**, **S3**
- Used **Databricks** as a platform for high-scaled analytics using spark.
- Worked on the Analytics Infrastructure team to develop a stream filtering system on top of **Apache Kafka**.
- Used **Python scripts** to update content in the database and manipulate files.
- Involved in supporting a **cloud-based data warehouse environment**. such as **Snowflake**
- Implemented code in **python** to retrieve and **manipulate data**.
- Worked with a data **governance team** to maintain data models, Metadata, Data Dictionaries define source fields and its definitions.
- Extensively involved in managing and reviewing **Hadoop log files**.
- Created monitors, alarms, notifications and logs for Lambda functions, **Glue Jobs**, **EC2 hosts** using **Cloudwatch**.
- Design and build **ETL pipelines** to automate ingestion of structured and unstructured data.
- Demonstrated a full understanding of the Fact/Dimension data warehouse design model, including **star** and **snowflake design methods**.
- Involved in file movements between **HDFS** and **AWS S3** and extensively worked with **S3 bucket** in **AWS**.
- Responsible for the design, implementation, and architecture of very large-scale data intelligence solutions around **Snowflake Data Warehouse**.
- Worked on **functions** in **Lambda** that aggregated the data from incoming events, and then stored result data in **Amazon DynamoDB**.
- Load and transformed large sets of structured, semi structured and unstructured data using **Spark** and **Databricks**.
- Used **Amazon EMR** for map reduction jobs and tested locally using **Jenkins**.
- Created Load Balancer on **AWS EC2** for stable cluster and services which provide fast and effective processing of Data.
- Developed **API** for using **AWS Lambda** to manage the servers and run the code in the **AWS** .
- Designed and implemented **RestApi** to access **snowflake DB platform**.
- Created different types of reports including Cross-tab, Conditional, and Drill-down, **OLAP** and Sub reports.
- Worked on **Oozie workflow** engine for job scheduling.
- Developed tools using **Python**, **Shell scripting**, **XML** to automate some of the menial tasks.
- Wrote many **T-SQL** Stored procedures to perform specific tasks as per the requirements.
- Presented data in a visually appealing tool **Tableau**.

Client: WIPRO LTD, Hyderabad, India

Oct 2018 – Sep 2020

Role: Data Analyst

Responsibilities:

- Gathered, managed and refined requirements from different modules, integrated key monitors including defects, parametric, rates, inventory, and cycle time in dashboard development for different modules.
- Extracted data from a variety of sources such as MySQL, Oracle 11g, MS SQL server etc. and analyzed large data sets using tools such as SQL, SAS, JMP.
- Managed project by leading team for successful delivery of an application by customizing, deploying Oracle SQL Pathfinder tool used for querying and joining data simultaneously from different databases.
- Performed data visualization for different modules using Tableau and ONE Click method. Worked with different teams on defining and representing key parameters etc.
- Wrote Python scripts to automate processes for identifying trends, outliers and anomalies in data.

- Expertise in data analysis, and finding root causes. Utilized statistical analysis tools such as JMP and SAS extensively for comparing data sets, identifying correlations, trends, analyzing data, determining fleet mismatches etc.
- Built scalable and deployable machine learning models.
- Extensively performed large data read/writes to and from csv and excel files using pandas. Tasked with maintaining RDD using SparkSQL.
- Utilized Sqoop to ingest real-time data. Used analytics libraries Sci-Kit Learn, MLLIB and MLxtend. Extensively used Python's multiple data science packages like Pandas, NumPy, matplotlib, Seaborn, SciPy, Scikit-learn and NLTK.
- Performed Exploratory Data Analysis, trying to find trends and clusters. Built models using techniques like Regression, Tree based ensemble methods, Time Series forecasting, KNN, Clustering and Isolation Forest methods.
- Worked on data that was a combination of unstructured and structured data from multiple sources and automated the cleaning using Python scripts.
- Extensively performed large data read/writes to and from csv and excel files using pandas. Tasked with maintaining RDD using SparkSQL.
- Communicated and coordinated with other departments to collect business requirements. Tackled highly imbalanced Fraud dataset using undersampling with ensemble methods, oversampling and cost sensitive algorithms.
- Improved fraud prediction performance by using random forest and gradient boosting for feature selection with Python Scikit-learn.

Environment: Spark, Hadoop, AWS, SAS Enterprise Guide, SAS/MACROS, SAS/ACCESS, SAS/STAT, SAS/SQL, ORACLE, MS-OFFICE, Python, Pandas, Numpy, Machine Learning (logistic regression), Gradient Descent algorithm, Bayesian optimization, Tableau.

Client: Syntel, Hyderabad, India

Jun 2016 –Sep 2018

Role: Data Analyst

Responsibilities:

- Performed **Data Analysis** using **SQL queries** on source systems to identify data discrepancies and determine data quality.
- Performed extensive **Data Validation, Data Verification** against **Data Warehouse** and performed debugging of the **SQL-Statements** and stored procedures for business scenarios.
- Designed and developed **Tableau dashboards** using stack bars, bar graphs, scattered plots, and Gantt charts.
- Familiar with **DBMS** table design, loading, Data Modeling, and experience in **SQL**.
- Worked on **ER/Studio** for **Conceptual, logical, physical data modeling** and for generation of **DDL scripts**
- Handled performance requirements for databases in **OLTP** and **OLAP models**.
- Analyzed the data which is using the maximum number of resources and made changes in the back-end code using **PL/SQL** stored procedures and triggers
- Performed data completeness, correctness, data transformation and **data quality testing** using **SQL**.
- Involved in designing **Business Objects** universes and creating reports.
- Conducted design walk through sessions with the **Business Intelligence team** to ensure that reporting requirements are met for the business.
- Prepared complex **T-SQL queries**, views and stored procedures to load data into staging area.
- Wrote **UNIX shell scripts** to invoke all the stored procedures, parse the data and load into flat files.
- Created reports analyzing large-scale databases utilizing **Microsoft Excel Analytics** within legacy systems.

Environment: SQL, PL/SQL, OLAP, OLTP, UNIX, MS Excel, T-SQL.

Education Details: -

- **Jawaharlal Nehru Technological University.(Hyderabad, India)**
Bachelor of Technology, Information Technology 2016

Aug 2012- Jun 2016